ANSWER KEY · AGENTIC HUNTING WORKSHOP

Module 2 — Search Your History (15 min, together)

> We do this together. Type along if you're hands-on; watch the screen if not. **Level coverage: L2.** You need your Module 1 hunts (H-0004, H-0005) on disk. You leave with a third: **H-0006**.

Keep the Splunk time picker on **Last 30 days**, same as Module 1.

Part A — Ask your own hunts (3 min)

Guiding question (orientation): *Before you search Splunk again: have you hunted anything like H-0004 before? Was WS-BUILD-01 the only host?*

Why: your own notes are the cheapest search you have. Ask them first.

```
athf similar --hunt H-0004
```

You should see **`No similar hunts found`**. That's expected: nothing in your history looks like H-0004 yet. We'll fix that by the end.

Now ask in plain words:

```
athf similar "post-install shell spawns recon"
```

You should see one hunt: **H-0004**, score about **0.40** ("Similar"). Widen the terminal if the columns are cut off.

> **Answer key — staff + self-study**
>
> Verified on a fresh `athf init` workspace (ATHF 0.24.0: PyPI, git HEAD and the pinned v0.24.0 tag) with M1's H-0004/H-0005 added, filled or not:
>
> - `--hunt H-0004` → `No similar hunts found`. The 3 examples are macOS / Linux cron / AWS Lambda. H-0005 scores 0.086, under the default 0.1 cut-off. If someone asks "is it broken?": `athf similar --hunt H-0004 --threshold 0` lists H-0005 0.086, then the examples at ~0.01.
> - Free text → H-0004 at 0.399 (0.365 once H-0006 exists). Exact score shifts a little with what attendees typed; ~0.4 is the point. Model answer (orientation Q): no. The only post-install-shell hunt you have is the one you just wrote, and it names one host. Your history can't tell you about other hosts yet. That's what Parts B and C find out, and H-0006 is what makes this command answer next time.

Part B — Search the indicator (3 min)

Guiding question (orientation): *H-0004's indicator is the package and version, `packmule-cli@2.4.1`. Search every log, every host. How many hosts do you expect?*

Why: this is the usual first move: take the exact thing you found and search it everywhere.

```
index=thrunt "packmule-cli@2.4.1"
| stats count by sourcetype host
```

There's no hostname in this search, and Last 30 days already covers the whole attack week. **Widening the time picker won't change the answer.**

Guiding question (interpretation): *One host. Does that mean WS-BUILD-01 was the only victim, or only that the indicator was only logged there?*

> **Answer key — staff + self-study**
>
> Expected: two rows, one host. `XmlWinEventLog · WS-BUILD-01.sudoshipping.local · 2` and `evidenceforge:ecar:json · WS-BUILD-01.sudoshipping.local · 2` (alice's `cmd.exe /c npm install -g packmule-cli@2.4.1`, as Sysmon/4688 and as eCAR process + file). The string appears on no other host, in any log source. (`host`, not `hostname`: only eCAR has `hostname`, so `by hostname` would hide the Windows event log rows.) One of the two XmlWinEventLog events is stamped Nov 12 23:33 UTC, so a run before then on "Last 30 days" shows 1 for that row. On class day it shows 2. Model answer (interpretation Q): only that the indicator was logged there. An indicator search finds more of what you already saw. The second host never logged this install line, so no amount of time or OR-clauses finds it. Dead-end to expect: someone tries `"*packmule*"`. That does hit the second host (its postinstall path is still `packmule-cli`), but also CI-RUNNER-03, the M4 reveal. Don't run it on screen. The point: you got lucky that a string carried over; the attacker picks strings. The behavior is what they can't change without changing the attack.

Part C — Search the behavior (5 min)

Guiding question (orientation): *Drop the hostname, the package and the command line. Keep only the shape from H-0004: node launches a shell. What do you expect?*

Why: an indicator is a string the attacker typed, and they can change it on the next box. The behavior is what the attack has to do, so it finds a host you have no indicator for.

C1. Where did node launch `cmd.exe`? Any host.

```
index=thrunt sourcetype="evidenceforge:ecar:json"
properties.parent_image_path="*\\node.exe" properties.image_path="*\\cmd.exe"
| table _time hostname principal pid ppid properties.parent_image_path properties.image_path properties.command_line
```

Guiding question (interpretation): *A second host appears. Fill in its column. Which strings changed? Which part of the search still matched?*

	WS-BUILD-01 (H-0004)	Second host
User	alice.chmod	?
Install line	`npm install -g packmule-cli@2.4.1`	?
node launches	`cmd.exe /c "packmule --activate-telemetry"`	?

> **Answer key — staff + self-study**
>
> Expected C1: exactly two rows, the only times node launches anything in the corpus.
>
> - `WS-BUILD-01` · alice.chmod · pid 5656 · ppid 5648 · `cmd.exe /c "packmule --activate-telemetry"` (M1's A2 dead end)
> - `WS-DEV-04` · carol.grep · pid 13664 · ppid 13648 · `cmd.exe /k "packmule --activate-telemetry"`, **~24h41m later** Both rows: parent `C:\Program Files\nodejs\node.exe`, child `C:\Windows\System32\cmd.exe`. The ppid is node's pid: 13648 on WS-DEV-04 is `node ...\packmule-cli\postinstall.js` (5648 on WS-BUILD-01), so the link is ppid = pid, same as Module 1. Filled table: second host is WS-DEV-04 · carol.grep · **no install line logged** · `cmd.exe /k` instead of `/c`. Every indicator differs, so B (and even a search for H-0004's exact activation command line) misses it. node → cmd.exe is the same, and C matched that. Measured on the class corpus (Nov data): postinstall 24h39m54s apart, node → shell 24h40m34s. Say "~25h".

C2. What did that shell run? Take the second host's `pid` from C1. Whatever that shell launched has it as its `ppid`, the same pivot as Module 1.

```
index=thrunt sourcetype="evidenceforge:ecar:json" ppid=<the cmd.exe pid on the second host>
| table _time hostname principal pid ppid properties.parent_image_path properties.image_path properties.command_line
```

Guiding question (interpretation): *Benign build step, or the same intrusion? What makes it more than a coincidence?*

> **Answer key — staff + self-study**
>
> **Type:** `ppid=13664` Expected C2: two rows, both WS-DEV-04 (pid 13664 is unique to that host; adding `hostname=WS-DEV-04` changes nothing). `whoami` (pid 13680) 13s after the shell, `systeminfo` (pid 13700) 40s after; both carol.grep, ppid 13664. Under 1m30s from node running `C:\Users\carol.grep\AppData\npm\packmule-cli\postinstall.js`. Model answer (interpretation Q): same behavior (node runs a post-install script, launches a shell, the shell runs host recon within ~2 min), same activation command, different host, user and day, and no install line with the H-0004 indicator. On WS-DEV-04 the recon comes straight out of node's shell (`/k` keeps it open); on WS-BUILD-01 it came from a sibling shell. The indicator search couldn't find it; the behavior written down in M1 did. Expect the question "does WS-DEV-04 beacon too?" It doesn't: the proxy shows no traffic from `10.60.10.14` to `cdn-packmule-telemetry.net` (only `.11` and CI-RUNNER-03's `.30.13`). That's a follow-up for H-0006, not something to chase now. Hint trigger: if C1 returns nothing, the paste doubled the backslashes (`\\\\node.exe` matches nothing). Exactly two, as printed. A single `\` or none still returns the two rows.

Part D — Write H-0006 (4 min)

Guiding question (decision): *What does the next hunter need to know: who the second victim is, or how you found them?*

Why: written down, this search is one `athf similar` away next time.

```
athf hunt new \
  --title "Same post-install shell + recon on a second host (behavior search)" \
  --technique T1195.002 \
  --tactic execution --tactic discovery \
  --platform windows \
  --data-source edr \
  --actor "CRON GOBLIN" \
  --hypothesis "CRON GOBLIN repeated the H-0004 behavior on another developer workstation: node launches a shell that runs host recon" \
  --behavior "node.exe launches cmd.exe, and a cmd.exe shell runs whoami/systeminfo/net/ipconfig: the H-0004 shape, any host" \
  --location "all hosts in index=thrunt (no hostname, no package name)" \
  --evidence "eCAR process create: parent node.exe -> cmd.exe; parent cmd.exe -> recon binaries" \
  --threat-context "H-0004's indicator (packmule-cli@2.4.1) matched only WS-BUILD-01; searched the behavior instead" \
  --hunter "<your-name>" \
  --non-interactive
```

Open the file. In a workspace with H-0004 and H-0005 this is **H-0006**. The command prints the path, `hunts/production/2026/Q4/H-0006.md`. Open it in your editor (macOS: `open -t hunts/production/2026/Q4/H-0006.md`). LEARN is already filled from the command.

Change the status, then paste CHECK and KEEP.

In the file	What to do
`status:` and **Status:**	`planning` → `completed`
CHECK	Paste the two **H-0006 CHECK** blocks
KEEP	Paste **H-0006 KEEP** and **H-0006 FOLLOW-UP**

H-0006 CHECK, queries. Paste over `[Your initial query]` and its fence lines.

```
~~~spl
index=thrunt "packmule-cli@2.4.1"
| stats count by sourcetype host

index=thrunt sourcetype="evidenceforge:ecar:json"
properties.parent_image_path="*\\node.exe" properties.image_path="*\\cmd.exe"
| table _time hostname principal pid ppid properties.parent_image_path properties.image_path properties.command_line

index=thrunt sourcetype="evidenceforge:ecar:json" ppid=<the cmd.exe pid on the second host>
| table _time hostname principal pid ppid properties.parent_image_path properties.image_path properties.command_line
~~~
```

H-0006 CHECK, notes. Paste over the `**Query Notes:**` line and its two bullets.

```
**Query Notes:**
- B: packmule-cli@2.4.1 only on WS-BUILD-01. Missed the second host.
- C1: node → cmd.exe: WS-BUILD-01 (/c), <the second host> (/k, +25h).
- C2: the shell on <the second host> ran whoami, systeminfo as <its user>.
```

H-0006 KEEP. Select from the `[Concise summary…]` line down to the second example row of the **Findings** table (the `suspected` one), and paste this over it.

```
**Suspected.** Second victim: <the second host> (<its user>).
Found by the behavior search; the indicator missed it.

### Findings

Things you're reporting as malicious. Only `confirmed` and `suspected` belong here.

| **Verdict** | **Subject** | **Ticket** | **Evidence** | **Independent Confirmation** |
|-------------|-------------|-----------|--------------|------------------------------|
| `suspected` | <the second host> · <its user> | N/A | node → cmd /k → recon, ~25h after WS-BUILD-01 | None — telemetry only |
```

H-0006 FOLLOW-UP. Under **Follow-up Actions**, replace the two example lines with:

```
- [ ] Does <the second host> beacon like WS-BUILD-01 (H-0005)?
```

Why `suspected`, not `confirmed`? Read the line under the Findings table: `confirmed` needs something from outside the logs. All you have is logs.

> **Answer key — staff + self-study**
>
> Type in the blocks: `<the second host>` = `WS-DEV-04`, `<its user>` = `carol.grep`, `<the cmd.exe pid on the second host>` = `13664`. Target shape: `HUNTS_EXPECTED/H-0006.md` (blocks with those values filled in). Check `status: completed`, the three queries with `ppid=13664`, one `suspected` Findings row for `WS-DEV-04` / `carol.grep`. Unreplaced placeholders still validate. Attendees who paste only the blocks still have template placeholders under OBSERVE, Query Performance, Ruled Out and Lessons Learned; `athf hunt validate` passes anyway. Hint trigger: a paste in the wrong place usually means the selection ran past `### Query Performance` or past the Findings table. Undo and reselect. Model answer (decision Q): both, but how you found them is what's reusable: "the H-0004 indicator matched one host; the behavior search found a second." The command's `--threat-context` already says it, which is why `athf similar` finds H-0006 below.

Stop here — you should see

```
athf hunt list  # H-0004, H-0005, H-0006, plus the 3 examples
athf hunt validate  # every hunt passes
athf similar --hunt H-0004  # now finds H-0006
```

Fell behind? Copy the `athf hunt new` command above, then the paste blocks, and you're caught up.

> **Answer key — staff + self-study**
>
> Verified: `athf hunt validate` → `6 valid, 0 invalid`. `athf similar --hunt H-0004` → `Found 1 similar hunts`, H-0006 at **~0.16** ("Somewhat similar"; 0.156–0.162 depending on what is filled in), well above the 0.1 cut-off. That's the payoff: the same command that found nothing in Part A now answers, because H-0006 is written down.

Next: H-0005 is still open — Module 3 starts there. You watch an agent do the research and the searching.